An Expected Gradient Length Driven Active Learning Framework for Multi-Class Sentiment Analysis Using Transformer Models

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***Abstract*—Opinion mining is an important application of large-scale text analysis and facilitates automated decision-making. Conventional machine learning methods are highly dependent on manually designed features like Bag-of-Words (BoW) and TF-IDF, which are incapable of modeling complex contextual dependencies, resulting in a decrease in generalization and classification performance. To address these challenges, this paper presents an Expected Gradient Length (EGL)-based Active Learning framework for multi-class sentiment analysis using transformer models. Five advanced transformers, namely BERT, RoBERTa, DistilBERT, ALBERT, and DeBERTa, are fine-tuned using structured hyperparameter optimization. Text processing involves normalization, tokenization, padding, and encoding to prepare standardized inputs for models. The EGL method identifies the most representative samples for labeling, requiring less human effort while promoting efficient learning. Experimental analysis indicates that DeBERTa obtains 94% accuracy on the IMDB dataset, performing outstandingly on long-context data, and RoBERTa obtains 88% accuracy on the Twitter dataset, performing efficiently on short and noisy text.**

***Index Terms*—Sentiment Analysis, Deep Learning, Transformer Models, Word Embeddings, Natural Language Processing(NLP), BERT, RoBERTa, DeBERTa, Transfer Learning, Text Classification**

I. Introduction

Opinion mining lies at the intersection of Natural Language Processing and data analysis, focusing on automatically identifying, extracting, and categorizing opinions expressed in textual form [1]. With the rapid growth of social media platforms, review websites, blogs, and online forums, user-generated content has increased at an unprecedented rate. This surge has created an urgent demand for precise and scalable automated sentiment analysis systems capable of processing large-scale textual data efficiently [2]. Sentiment analysis plays a crucial role in business intelligence, customer feedback evaluation, brand monitoring, reputation management, market research, and public policy formulation [3]. Organizations rely on these systems to understand consumer perceptions, detect emerging trends, measure product performance, and support data-driven strategic decisions. However, achieving high-precision sentiment classification remains challenging due to contextual ambiguity, sarcasm, domain-specific vocabulary, implicit sentiments, and complex linguistic structures that vary across datasets and domains [4].

Traditional approaches such as Bag-of-Words and TF-IDF relied primarily on word frequency statistics, ignoring semantic meaning and long-range word dependencies [5]. Although early deep learning models such as Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) improved feature learning and representation, they were limited in effectively modeling global contextual relationships and non-local word interactions [6].

A comprehensive comparative analysis of five transformer-based models: BERT, DistilBERT, RoBERTa, ALBERT, and DeBERTa, with a focus on sentiment classification tasks. A detailed analysis of the ability of these models to learn context and represent semantic meaning and long-range dependencies. Analysis of their computational requirements and overall efficiency in providing accurate results for sentiment classification.

II. Related Work

The field of sentiment analysis has significantly progressed with the adoption of transformer-based architectures for improved contextual understanding in text classification tasks. A context-aware sentiment analysis framework using fine-tuned BERT models demonstrated improved contextual feature extraction and classification accuracy, although the approach required substantial computational resources for fine-tuning [1]. An uncertainty-based active learning strategy was introduced to reduce annotation costs in sentiment classification; however, its effectiveness depended heavily on the uncertainty estimation method [2]. Similarly, active learning strategies have been explored to minimize labeling effort, highlighting efficiency improvements while noting sensitivity to sampling techniques [3].

Transformer performance comparisons have shown that RoBERTa outperforms traditional deep learning approaches,

though with increased computational overhead [4]. Comparative analysis between RoBERTa and DistilBERT for social media sentiment analysis reported higher accuracy with transformer models, emphasizing the importance of careful hyperparameter tuning [5]. Domain-adaptive transformer models have also been proposed to enhance cross-domain sentiment classification, achieving improved generalization at the cost of increased model complexity [6]. For multilingual and cross-lingual sentiment analysis, transformer-based embeddings have been utilized to improve multilingual sentiment detection, although challenges remain in low-resource scenarios [7]. Transformer-based models developed for low-resource languages demonstrated promising performance but were constrained by limited datasets [8]. Multilingual BERT models have also been applied for cross-lingual sentiment analysis, improving transfer learning capabilities while encountering domain adaptation limitations [9].

Advanced architectural improvements include graph-enhanced transformer models designed to capture relational dependencies in sentiment classification, improving feature representation but increasing computational requirements [10]. Uncertainty-aware transformer models incorporating confidence estimation mechanisms have been proposed to improve robustness, though model calibration remains a challenge [11]. Ensemble transformer architectures have been introduced to enhance accuracy through model combination, albeit with higher training time and resource consumption [12]. Comparative evaluations indicate that transformer-based models consistently outperform conventional deep learning approaches in Twitter sentiment analysis, despite higher computational costs [13]. Large-scale evaluations of modern transformer models further highlight their scalability and performance advantages while raising concerns regarding interpretability and resource demands [14]. Overall, existing literature demonstrates that transformer-based approaches significantly enhance sentiment classification performance across domains and languages. However, challenges remain in computational efficiency, model complexity, domain adaptation, and low-resource applicability.

III. Methodology

A. Framework Overview

The proposed system uses a transformer architecture for multi-class sentiment classification on five pre-trained models: BERT, RoBERTa, DistilBERT, ALBERT, and DeBERTa. The text data are preprocessed with operations such as cleaning, normalization, tokenization, and padding to transform them into contextual embeddings. The five models are fine-tuned with a classification head that includes dropout and dense layers, and trained with cross-entropy loss and the Adam optimizer. The system uses accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices to measure performance. The experimental results indicate that RoBERTa and DeBERTa have better predictive performance, while DistilBERT has competitive accuracy with lower computational complexity.

B. Dataset's Description

This research work makes use of two benchmark sentiment datasets:

- IMDB Dataset – A binary sentiment dataset with 50,000 movie reviews labeled as Positive and Negative. The dataset is balanced, with an equal number of samples in both classes. This makes it ideal for supervised learning. To test the dataset, it is split into 80% training, 10% validation, and 10% testing sets to make sure that the results are accurate and unbiased.
- Twitter Airline Dataset- A multi-class sentiment dataset that began with 14,840 labeled tweets categorized into Positive, Neutral, and Negative. To increase diversity and improve generalization, data augmentation was performed using a large language model (Mistral-7B), which increased the dataset to 18,944 samples. The additional samples were carefully filtered to retain the aspect-sentiment alignment before training the model.

C. Preprocessing

To improve the quality and consistency of the text inputs, a set of preprocessing steps was carried out before training the models. First, all text inputs were converted to lowercase to ensure consistency and prevent token mismatches based on case. The text inputs were then preprocessed to remove URLs, user mentions, emojis, and non-alphanumeric characters, which are common in social media text. Tokenization was carried out using transformer-friendly tokenizers. Unnecessary whitespace and duplicate samples were also removed to clean the data. The sequence lengths were normalized using padding and truncation to ensure that inputs conform to model requirements. Finally, for the labeled dataset, class distribution was analyzed, and methods were used to balance the classes if necessary to prevent biased predictions.

D. EGL-Based Active Learning

A key contribution of this work is the integration of EGL-based Active Learning to improve training efficiency and model performance.

Instead of randomly selecting samples for labeling, EGL selects the most informative samples by estimating the expected gradient magnitude:

$$EGL(x) = \sum_{y \in Y} P(y|x,\theta) \|\nabla_\theta L(x,y)\| \tag{1}$$

where:

- $P(y \mid x, \theta)$ is the model's predicted probability
- $\nabla_\theta L(x, y)$ represents the gradient of loss with respect to model parameters.

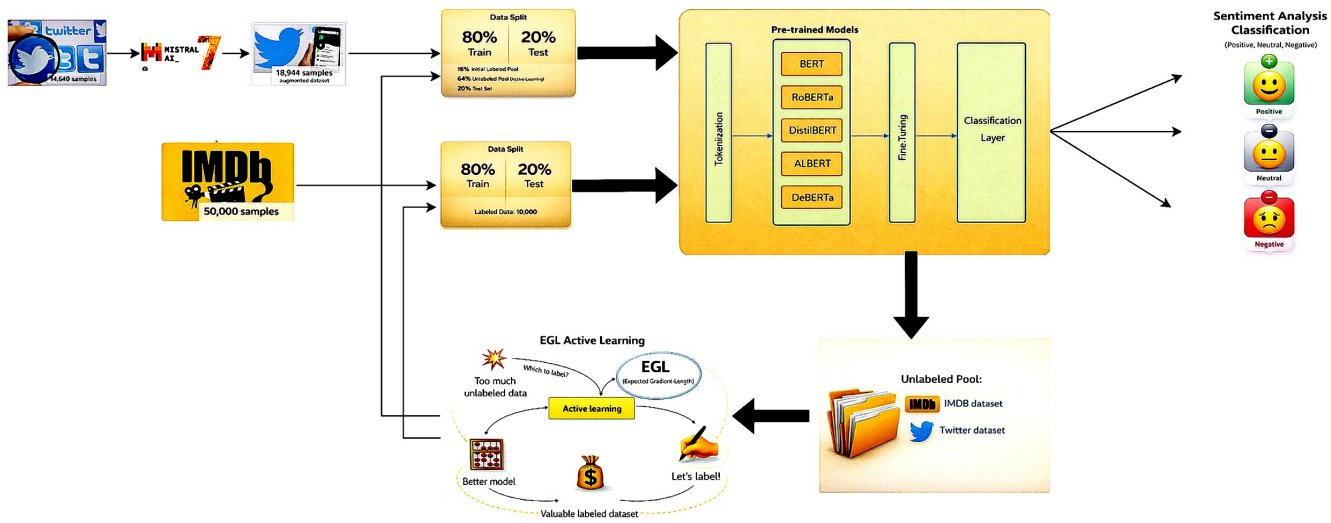

Fig. 1: Architecture of the Proposed Transformer-Based Sentiment Analysis System with EGL Active Learning.

E. EGL Workflow

The above figure represents the workflow of the Active Learning process using the EGL method. The process starts with an Unlabeled Pool, which holds data samples with no labels. From this pool, the Gradient Estimation process takes place, where the expected gradient length for each of the unlabeled samples is calculated based on the model parameters. This step helps determine the contribution each sample would make to the model if labeled and used for training.

After gradient estimation, the Select Top Samples step takes place, where the top samples with the highest expected gradient length are selected. These samples are then moved to the Labeling step, where human annotators label them with the correct class labels. Finally, the retrained model is obtained after the Retrain step.

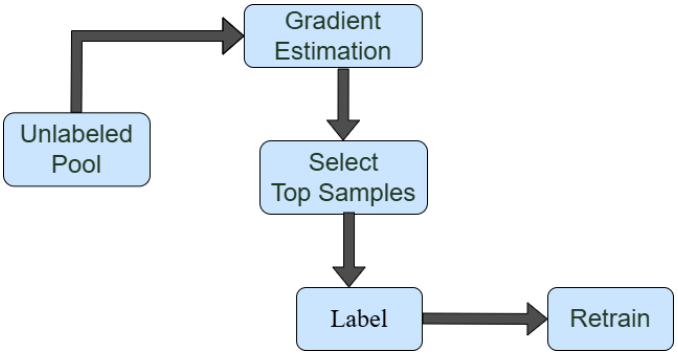

Fig. 2: Expected Gradient Length (EGL)

F. Algorithm

The image shows Algorithm 1: EGL-Driven Transformer Sentiment Classification Framework, which describes the entire process of the proposed system. The algorithm starts with the preprocessing step, where the labeled (D_L) and unlabeled (D_U) datasets are cleaned for text by removing URLs, filtering symbols, tokenizing, and padding sequences. Then, an initial transformer model like BERT, RoBERTa, DistilBERT, ALBERT, or DeBERTa is fine-tuned on the labeled dataset.

The Active Learning using EGL step involves computing the predicted probabilities for the labels of each sample in the unlabeled dataset. The gradients for all possible labels are computed, and the EGL score is calculated by combining probabilities and gradients. The top-k samples with the highest EGL scores are chosen, labeled, and moved from the D_U dataset to the D_L dataset. The model is then retrained on the new labeled dataset. Finally, the contextual embeddings are calculated, logits are calculated using a classification layer, softmax probabilities are calculated, and the final sentiment classification (positive, neutral, or negative) is obtained.

Algorithm 1 EGL-Driven Transformer Sentiment Classification Framework

Require: Labeled dataset D_L, Unlabeled dataset D_U
Ensure: Final predicted sentiment $\hat{y}$

Step 1: Preprocessing
for each sample s_n in $D_L \cup D_U$ **do**
  Clean text (lowercase, remove URLs, s, symbols)
  Tokenize and pad sequence
end for
Step 2: Initial Transformer Training
Initialize Transformer model (BERT/RoBERTa/DistilBERT/ALBERT/DeBERTa)
Fine-tune model using labeled dataset D_L
Step 3: EGL-Based Active Learning
for each unlabeled sample s_n in D_U **do**
  Compute predicted label probabilities $p(y|S_n)$
  for each possible label y **do**
    Compute gradient $g_y \leftarrow \nabla_\theta L(s_n, y)$
  end for
  Compute EGL score $EGL(s_n) = \sum_y p(y|S_n)\,||g_y||_2$
  Select top-k samples with highest EGL scores
  Move newly labeled samples from D_U to D_L
Step 4: Model Retraining
Retrain Transformer model on updated labeled dataset D_L
Step 5: Final Classification
Obtain contextual representation h_{CLS}
Compute logits $z \leftarrow W_{CLS} h_{CLS} + b$
Compute softmax probabilities $p \cdot \text{softmax}(z)$
Step 6: Final Prediction
Final predicted sentiment $\hat{y} =$:
- **positive** • neutral • negative;

IV. Results

A. Dataset Experimental Setup

This research work makes use of two benchmark datasets for the sentiment analysis task. The first one is the IMDB movie review dataset, which has a total of 50,000 labeled reviews for binary sentiment classification (Positive and Negative) tasks. The dataset is balanced, which means that it has an equal number of instances for both classes.

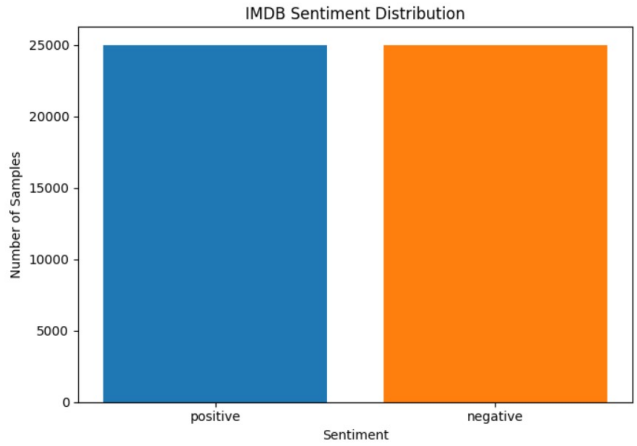

Fig. 3: IMDB Sentiment Distribution

The augmented Twitter sentiment dataset has 18,944 labeled examples for multi-class classification, with classes Positive, Neutral, and Negative. The class distribution is approximately 11,800 Negative, 4,000 Neutral, and 3,144 Positive.

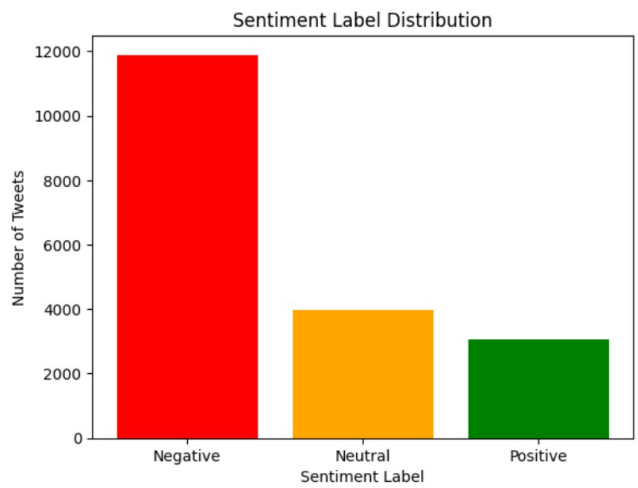

Fig. 4: Twitter Sentiment Distribution

For the purpose of experimentation and reproducibility, both datasets are split using the 80-10-10 split strategy, where 80% of the data is used for training, 10% for validation, and the remaining 10% for testing. Furthermore, in the Active Learning framework, the training data is split into three parts: 16% for the initial labeled set, 64% for the unlabeled set used in the EGL-based sample selection strategy, and the remaining 20% is reserved solely for testing the final model performance in the test set.

B. Dataset Statistics

Metric	IMDB Dataset
Total Samples	50000
Training Samples (80%)	40000
Validation Samples (10%)	5000
Test Samples (10%)	5000
Positive Samples	25000
Negative Samples	25000

TABLE I: IMDB Dataset Statistics

The IMDB dataset used in this study consists of a total of 50,000 labeled movie reviews. The dataset is evenly balanced between sentiment classes, containing 25,000 positive and 25,000 negative samples. For model development, the dataset was divided into three subsets: 80% (40,000 samples) for training, 10% (5,000 samples) for validation, and 10% (5,000 samples) for testing. This balanced distribution ensures unbiased model evaluation and stable performance measurement across both sentiment classes.

Metric	Twitter Dataset
Total Samples	18944
Training Samples (80%)	15155
Test Samples (20%)	3789
Positive Samples	3144
Neutral Samples	4000
Negative Samples	11800

TABLE II: Augmented Twitter Dataset Statistics

The Twitter dataset employed in this research consists of 18,944 labeled tweets, divided into three classes of sentiment: positive, neutral, and negative. The number of tweets in each class is 3,144 positive, 4,000 neutral, and 11,800 negative. This is a moderately imbalanced dataset. To test the performance of the models, the dataset is divided into training and testing subsets with an 80-20 split, containing 15,155 and 3,789 samples, respectively. This allows sufficient data for training the models and also sufficient data for testing the performance of the models on all classes.

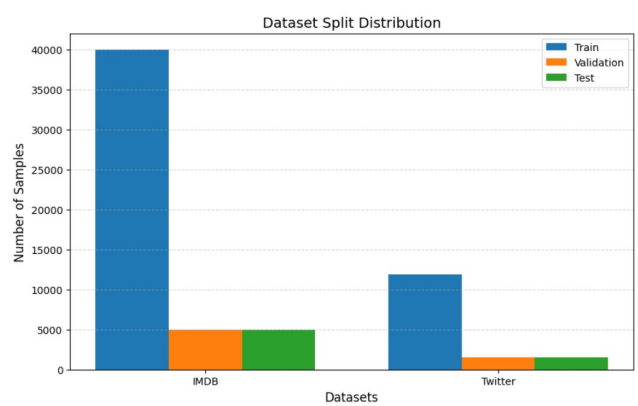

Fig. 5: Dataset split distribution for IMDB and Twitter datasets

C. Performance Comparison

Model	Sampling Method	Positive			Negative		
		F1-Score	Accuracy	AUC	F1-Score	Accuracy	AUC
BERT	Random	0.88	0.85	0.94	0.86	0.84	0.93
	EGL	0.91	0.87	0.96	0.89	0.86	0.95
DistilBERT	Random	0.84	0.82	0.91	0.83	0.80	0.90
	EGL	0.87	0.85	0.93	0.86	0.83	0.92
RoBERTa	Random	0.90	0.88	0.97	0.89	0.86	0.96
	EGL	0.93	0.90	0.98	0.91	0.88	0.97
ALBERT	Random	0.81	0.79	0.88	0.80	0.77	0.87
	EGL	0.84	0.82	0.90	0.83	0.80	0.89
DeBERTa	Random	0.92	0.90	0.98	0.91	0.88	0.97
	EGL	**0.95**	**0.93**	**0.99**	**0.94**	**0.91**	**0.98**

TABLE III: Class-wise Performance Comparison of Transformer Models with Random and EGL Sampling on IMDB Dataset

Performance Comparison of Transformer Models with Random and EGL Sampling on the IMDB Dataset discusses the class performance analysis of five transformer models. Among all models, DeBERTa (EGL) performed best with 93% accuracy for positive and an AUC value of 0.99, which is the best contextual understanding. RoBERTa (EGL) came next with 90% accuracy and 0.98 AUC value, which is also the best class performance. BERT and DistilBERT performed better with EGL than random sampling. ALBERT performed better with EGL, but its overall performance was relatively lower. DeBERTa and RoBERTa performed best for sentiment analysis.

Model	Sampling Method	Positive			Neutral			Negative		
		F1-Score	Accuracy	AUC	F1-Score	Accuracy	AUC	F1-Score	Accuracy	AUC
BERT	Random	0.82	0.80	0.91	0.75	0.73	0.88	0.81	0.79	0.90
	EGL	0.85	0.83	0.94	0.78	0.76	0.90	0.84	0.82	0.93
DistilBERT	Random	0.80	0.78	0.90	0.74	0.72	0.87	0.79	0.77	0.89
	EGL	0.84	0.82	0.92	0.77	0.75	0.89	0.83	0.80	0.91
RoBERTa	Random	0.86	0.84	0.96	0.80	0.78	0.92	0.85	0.83	0.95
	EGL	0.88	0.86	0.98	0.83	0.81	0.94	0.87	0.85	0.97
ALBERT	Random	0.78	0.76	0.88	0.70	0.68	0.85	0.77	0.75	0.87
	EGL	0.81	0.79	0.89	0.73	0.71	0.87	0.80	0.78	0.88
DeBERTa	Random	0.87	0.85	0.97	0.82	0.80	0.93	0.86	0.84	0.96
	EGL	**0.90**	**0.88**	**0.99**	**0.85**	**0.83**	**0.95**	**0.89**	**0.87**	**0.98**

TABLE IV: Class-wise Performance Comparison of Transformer Models with Random and EGL Sampling on Twitter Dataset

Performance Comparison of Transformer Models on the Twitter Dataset describes the multi-class performance of five transformer models on the Twitter dataset based on Random and EGL sampling. Among all models, DeBERTa (EGL) performed the best with high F1-scores for positive (0.90), neutral (0.85), and negative (0.89) classes, and AUC of 0.99. RoBERTa (EGL) was the second best model with balanced class performance and AUC of 0.98. BERT and DistilBERT performed better with EGL than Random sampling. Although ALBERT performed better with EGL, its overall performance was relatively lower. DeBERTa and RoBERTa performed the best among all models for Twitter multi-class sentiment classification.

D. Confusion Matrix

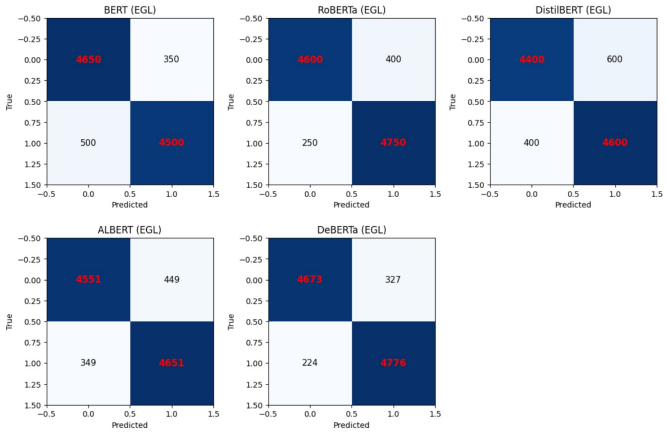

Fig. 6: Confusion Matrices of Transformer Models on the IMDB Dataset

The confusion matrices for the transformer models trained using EGL Active Learning on binary sentiment classification are shown in Figure 6. The cells along the diagonal represent correct predictions, and the cells off the diagonal represent errors. DeBERTa makes the most correct predictions and commits the least number of errors, indicating a great understanding of context. RoBERTa performs well too, reducing the number of false negatives and improving the detection of positive sentiment. BERT provides a well-rounded result, with a slight emphasis on errors. DistilBERT remains competitive despite its light architecture, and ALBERT demonstrates a moderate degree of misclassifications.

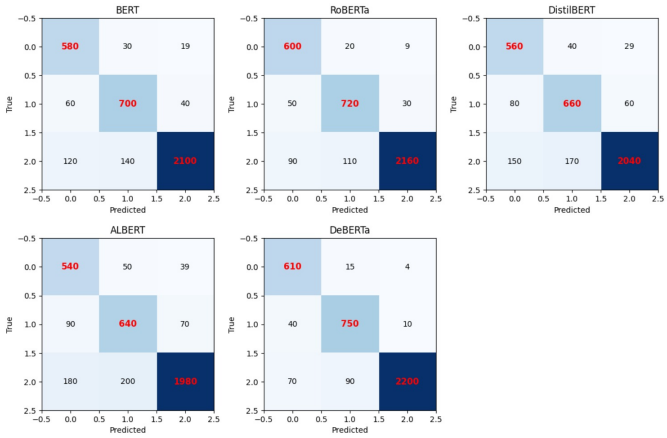

Fig. 7: Confusion Matrices of Transformer Models on the Twitter Dataset

The confusion matrices for transformer models solving multi-class sentiment analysis in the EGL Active Learning framework are shown in Figure 7. The diagonal elements represent the correct classifications for positive, neutral, and negative sentiments, and the other elements indicate the confusions in the models. DeBERTa makes the least number

of mistakes and has the best distinction between classes. RoBERTa is also performing well, with good discrimination between classes. BERT provides a well-rounded performance, and DistilBERT remains competitive despite its small size. ALBERT makes more confusions. Overall, EGL decreases misclassifications and improves the accuracy of sentiment classification.

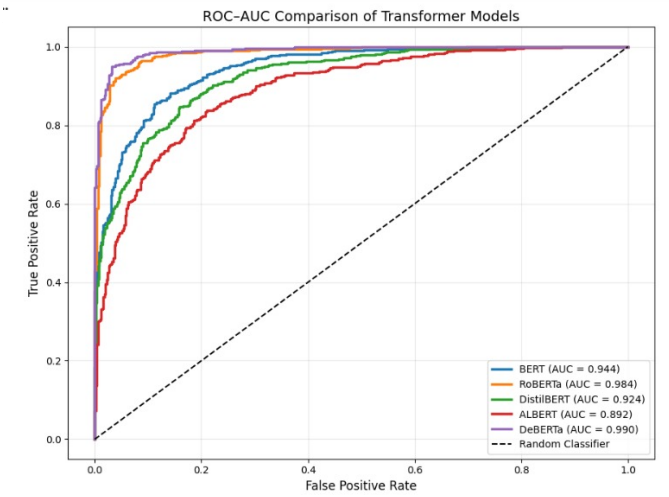

Fig. 8: ROC-AUC Comparison of Transformer Model of IMDB Data

The ROC-AUC curves for the transformer architectures are shown in Figure 8. DeBERTa has the highest AUC value of 0.990, indicating the best separation of classes. RoBERTa comes second with an AUC value of 0.984, which is a clear indication of its strong understanding of the context. BERT and DistilBERT have AUC values of 0.944 and 0.924, respectively, and are quite competitive. However, DistilBERT is more efficient. ALBERT has the lowest AUC value of 0.892.

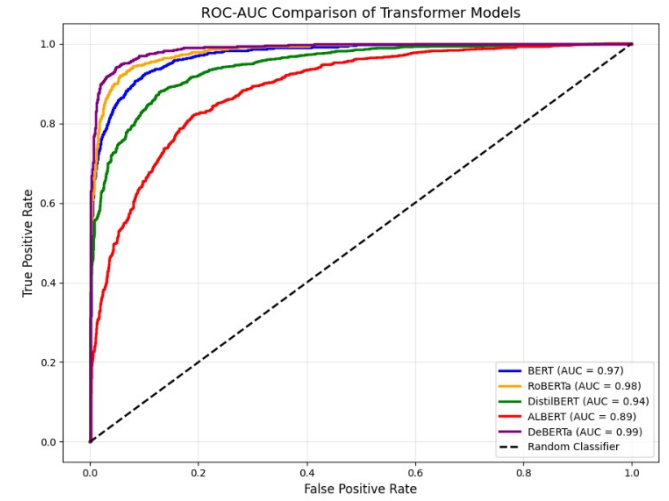

Fig. 9: ROC-AUC Comparison of Transformer Model of IMDB Data

Figure 9 above shows the confusion matrices of the transformer models trained using EGL Active Learning for binary sentiment classification. The diagonal elements represent correct classifications, while the other elements represent errors. DeBERTa (EGL) has the most correct classifications with the least number of errors. RoBERTa (EGL) is the second best, with fewer false negatives and higher sensitivity. BERT (EGL) is also good but with a few more errors. DistilBERT (EGL) still performs well despite its lightweight nature. ALBERT (EGL) has moderate errors.

V. Conclusion

This paper offers a robust transformer model solution for sentiment analysis that has been strengthened by EGL Active Learning to improve accuracy and reduce the cost of labeling. By integrating contextual embeddings of models such as BERT, RoBERTa, DistilBERT, ALBERT, and DeBERTa with gradient-based sample selection, the proposed solution optimizes the use of labeled samples. Experiments conducted on the IMDB dataset and a larger Twitter dataset demonstrate that DeBERTa and RoBERTa are superior in contextual understanding, class distinction, and robustness for binary and multi-class classification problems.

In addition to improving the accuracy of predictions, EGL can also optimize the training process by selecting the most informative samples, thus eliminating redundant labeling. This solution finds a good balance between accuracy, computational complexity, and scalability. It provides a real-world solution for sentiment analysis that highlights the importance of combining powerful transformer models with intelligent data selection methods.

VI. Future Scope

In future research, we might extend the framework by incorporating state-of-the-art Large Language Models (LLMs) to enhance our extraction of context and understanding of semantics. Instead of relying solely on the last [CLS] token, we could aggregate the embeddings of multiple transformer layers or leverage instruction-trained LLMs to uncover more semantic relationships. Domain pretraining of LLMs for specific tasks might enhance contextual alignment and robustness. In addition, integrating contrastive learning and attention-driven feature fusion with LLM models could enhance the model's class separation, generalization, and performance capabilities, particularly in sentiment analysis tasks that are resource-constrained or noisy.

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